Force Estimation, Realtime Filtering, and Leak-Free Uncertainty Bands

HSMR 2024 Effectiveness Force-Estimation Pipeline

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1 Purpose

The goal is to estimate external wrench from motor torques and the robot model, then attach a confidence interval (CI) without using the force sensor to tune or compute that interval. The force sensor is used only for validation plots and RMSE/coverage checks.

> **No knowledge leakage rule.** The measured force sensor signal is never used inside the uncertainty model. It is only used after estimation to compare against $\hat{\boldsymbol{F}}$.

2 Signals and Main Force Equation

The LSTM predicts free-space joint torque. The measured motor torque contains free-space dynamics plus external-contact effects. The torque residual is

$$\boldsymbol{y}_k = \tau_{\text{meas}}, k - \tau_{\text{LSTM}}, k. \tag{1}$$

Ideally,

$$\boldsymbol{y}_k \approx \tau_{\text{ext}}, k. \tag{2}$$

Given a filtered external torque estimate $\hat{\boldsymbol{\tau}}_{\text{ext},k}$, the Cartesian wrench is computed using the Jacobian transpose relationship:

$$\hat{\boldsymbol{\tau}}_{\text{ext},k} = J_k^{\mathsf{T}} \hat{\boldsymbol{F}}_k, \tag{3}$$

$$\hat{\boldsymbol{F}}_k = J_k^{-\mathsf{T}} \hat{\boldsymbol{\tau}}_{\text{ext},k}. \tag{4}$$

The first three force channels are then rotated into the plotting frame using the same rotation used in the original MATLAB script:

$$\hat{\boldsymbol{F}}_{k,1:3}^{\text{plot}} = R \hat{\boldsymbol{F}}_{k,1:3}. \tag{5}$$

3 Realtime Estimator Variants

All estimators operate in joint-torque space first. The force projection $\hat{\boldsymbol{F}} = J^{-\mathsf{T}} \hat{\boldsymbol{\tau}}_{\text{ext}}$ is applied after filtering.

3.1 Direct Residual

The simplest estimator uses the raw torque residual:

$$\hat{\boldsymbol{\tau}}_{\mathrm{ext},k} = \boldsymbol{y}_k. \tag{6}$$

This is responsive but noisy and tends to preserve free-space false positives.

3.2 Paper-Style Residual Observer

The observer form discussed in the paper is

$$\dot{\boldsymbol{r}} = K_i(\hat{\tau}_{\mathrm{ext}} - \boldsymbol{r}). \tag{7}$$

In discrete time, the implementation uses an exact first-order update:

$$\boldsymbol{r}_k = \alpha_k \boldsymbol{r}_{k-1} + (1-\alpha_k)\boldsymbol{y}_k, \tag{8}$$

$$\alpha_k = \exp(-K_i \Delta t_k). \tag{9}$$

The force estimate uses

$$\hat{\boldsymbol{\tau}}_{\mathrm{ext},k} = \boldsymbol{r}_k. \tag{10}$$

This is realtime and stable. Larger K_i gives faster tracking but less smoothing. The current tuned value is

$$K_i = 15\ \mathrm{s}^{-1}. \tag{11}$$

3.3 Bias-Gated Observer

The bias-gated observer tries to avoid false contact detection by estimating a slow free-space bias only when the residual looks close to zero-contact:

$$\boldsymbol{b}_k \approx \text{slow bias in } \boldsymbol{y}_k. \tag{12}$$

The corrected residual is

$$\tilde{\boldsymbol{y}}_k = \mathrm{deadband}(\boldsymbol{y}_k - \boldsymbol{b}_k). \tag{13}$$

Then the observer is applied to $\tilde{\boldsymbol{y}}_k$. In our tests, nonzero bias adaptation often worsened the result, so the practical setting has been

$$\text{bias gain} = 0. \tag{14}$$

3.4 Two-Time-Scale Observer

This method subtracts a slow free-space estimate from a fast observer:

$$\hat{\boldsymbol{\tau}}_{\mathrm{ext},k} = \mathrm{deadband}(\boldsymbol{r}_k^{\mathrm{fast}} - \boldsymbol{r}_k^{\mathrm{slow}}). \tag{15}$$

The idea is sensible, but in the current data it did not clearly outperform the simpler tuned observer or the smooth-deadband methods.

3.5 Smooth Torque-Space Deadband

The most useful false-positive suppression came from applying a smooth per-joint deadband before filtering. For joint i,

$$d_i = m\, s_i\, \sigma_{\tau,i}, \tag{16}$$

where m is the global deadband multiplier, s_i is a joint-specific multiplier, and $\sigma_{\tau,i}$ is estimated from the available torque residual signal only.

The smooth gate is

$$g_i(y) = \frac{1}{1 + \exp\left(-\frac{|y|-d_i}{w_i}\right)}, \tag{17}$$

$$\tilde{y}_{i,k} = y_{i,k} g_i(y_{i,k}). \tag{18}$$

For the dVRK-Si robot, joints 1–2 are noisier and joint 3 tends to underestimate in free space, so the current joint multipliers are

$$\boldsymbol{s} = [1,\ 1,\ 0.25,\ 1,\ 1,\ 1]. \tag{19}$$

3.6 Kalman Filter in Torque Space

The Kalman filter is implemented before the force projection. The state is the external joint torque:

$$\boldsymbol{x}_k = \tau_{\text{ext}}, k. \tag{20}$$

The process model is a random walk:

$$\boldsymbol{x}_k = \boldsymbol{x}_{k-1} + \boldsymbol{w}_k, \qquad \boldsymbol{w}_k \sim \mathcal{N}(0, Q). \tag{21}$$

The measurement is the LSTM torque residual:

$$\boldsymbol{y}_k = \tau_{\text{meas}}, k - \tau_{\text{LSTM}}, k = \boldsymbol{x}_k + \boldsymbol{v}_k, \qquad \boldsymbol{v}_k \sim \mathcal{N}(0, R). \tag{22}$$

For each joint channel, the realtime Kalman recursion is

$$P_{k|k-1} = P_{k-1|k-1} + Q, \tag{23}$$

$$K_k = \frac{P_{k|k-1}}{P_{k|k-1} + R}, \tag{24}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - \hat{x}_{k|k-1}), \tag{25}$$

$$P_{k|k} = (1 - K_k)P_{k|k-1}. \tag{26}$$

The useful variant is not the raw Kalman filter alone, but

`kalman_smooth_deadband`,

which first applies the smooth torque-space deadband and then applies the Kalman filter. This keeps the false-positive suppression and replaces the fixed observer gain with a probabilistic filter.

4 Uncertainty Model

The CI is leak-free: it uses motor torque residuals, the Jacobian, the estimated force, and the filter disagreement. It does not use measured force.

4.1 Torque-Side Noise Estimate

Torque residual noise is estimated from first differences:

$$\Delta \boldsymbol{y}_k = \boldsymbol{y}_k - \boldsymbol{y}_{k-1}. \tag{27}$$

The per-joint robust standard deviation is

$$\sigma_{\tau,i} \approx \frac{1.4826 \text{ median} \left(|\Delta y_i - \text{median}(\Delta y_i)| \right)}{\sqrt{2}}. \tag{28}$$

The factor $\sqrt{2}$ appears because differencing two independent noisy samples doubles the variance.

4.2 Propagating Torque Uncertainty to Force

At each time step,

$$A_k = R_6 J_k^{-\mathsf{T}}, \tag{29}$$

where R_6 applies the force-axis rotation to the first three wrench components. The propagated wrench variance is

$$\boldsymbol{\sigma}^2_{\text{prop},k} = A_k^{\circ 2} \boldsymbol{\sigma}_\tau^2, \tag{30}$$

where $A_k^{\circ 2}$ means elementwise squaring.

4.3 Low-Force Ambiguity

Near zero contact, the distinction between no contact, friction disturbance, and small external force is ambiguous. This is modeled with

$$s_F(|\hat{\boldsymbol{F}}_k|) = 1 + g_F \exp\left(-\frac{|\hat{\boldsymbol{F}}_k|}{F_0} \right). \tag{31}$$

Current values:

$$g_F = 3.0, \qquad F_0 = 10 \text{ N}. \tag{32}$$

4.4 Observer/Direct Disagreement

The direct projected wrench is

$$\hat{\boldsymbol{F}}_k^{\text{direct}} = J_k^{-\mathsf{T}} \boldsymbol{y}_k. \tag{33}$$

The filtered estimate is

$$\hat{\boldsymbol{F}}_k^{\text{filtered}} = J_k^{-\mathsf{T}} \hat{\boldsymbol{\tau}}_{\text{ext},k}. \tag{34}$$

Their disagreement is used as a model-risk term:

$$\boldsymbol{\sigma}_{\text{disagree},k} = g_D \left| \hat{\boldsymbol{F}}_k^{\text{direct}} - \hat{\boldsymbol{F}}_k^{\text{filtered}} \right|. \tag{35}$$

Current value:

$$g_D = 0.8. \tag{36}$$

4.5 Relative Force Uncertainty

Large forces should not become unrealistically certain. A relative calibration term is added:

$$\boldsymbol{h}_{\text{rel},k} = g_R |\hat{\boldsymbol{F}}_{k,1:3}|. \tag{37}$$

Current value:

$$g_R = 0.10. \tag{38}$$

The F_z channel also receives an empirical dVRK-Si axis correction:

$$h_{z,k} \leftarrow g_z h_{z,k}, \qquad g_z = 2.0. \tag{39}$$

4.6 Final 95% Band

The final standard deviation estimate is

$$\boldsymbol{\sigma}_{F,k} = \sqrt{\left(s_F(|\hat{\boldsymbol{F}}_k|) \boldsymbol{\sigma}_{\text{prop},k} \right)^2 + \boldsymbol{\sigma}^2_{\text{disagree},k}}. \tag{40}$$

The 95% half-width is

$$\boldsymbol{h}_k = 1.96\, \boldsymbol{\sigma}_{F,k}. \tag{41}$$

For the first three force channels, the relative term is combined as

$$\boldsymbol{h}_{k,1:3} \leftarrow \sqrt{\boldsymbol{h}^2_{k,1:3} + \left(g_R |\hat{\boldsymbol{F}}_{k,1:3}| \right)^2}. \tag{42}$$

Then the plotted confidence interval is

$$\hat{\boldsymbol{F}}_k - \boldsymbol{h}_k \leq \boldsymbol{F}_k \leq \hat{\boldsymbol{F}}_k + \boldsymbol{h}_k. \tag{43}$$

A maximum plotted half-width is used for readability:

$$h_{i,k} \leq 12 \text{ N}. \tag{44}$$

5 Current Key Hyperparameters

Parameter	Current value	Meaning
K_i	15 s^{-1}	Observer speed
Deadband multiplier	20	Torque residual suppression
Joint deadband multipliers	$[1, 1, 0.25, 1, 1, 1]$	dVRK-Si joint-specific correction
Kalman process scale	0.08	Higher tracks faster but is noisier
Kalman measurement scale	1.0	Higher trusts residual less
g_F	3.0	Low-force uncertainty inflation
F_0	10 N	Low-force decay scale
g_D	0.8	Observer/direct disagreement uncertainty
g_R	0.10	Relative force-scale uncertainty
g_z	2.0	F_z axis uncertainty correction

6 Validation Summary

On the current dataset, the useful Kalman variant is

`kalman_smooth_deadband`.

Pure Kalman filtering of the raw residual is worse, because the raw residual still contains free-space false positives and low-velocity friction/model disturbances. The smooth deadband remains important.

Method	F_x RMSE	F_y RMSE	F_z RMSE	Mean
Raw Kalman, $q = 0.08$	2.4528	2.3832	1.4578	2.0979
Smooth-deadband observer	1.6700	1.6260	1.4385	1.5782
Smooth-deadband Kalman, $q = 0.08$	1.6800	1.6266	1.4257	1.5775

Interpretation: the Kalman-smooth method is essentially tied with the tuned observer overall, slightly improves F_z, and slightly worsens F_x. Its value is that the filter is more principled and tunable through Q/R, while remaining realtime.

7 Plot Explanations

7.1 Force Estimate With 95% Confidence Bands

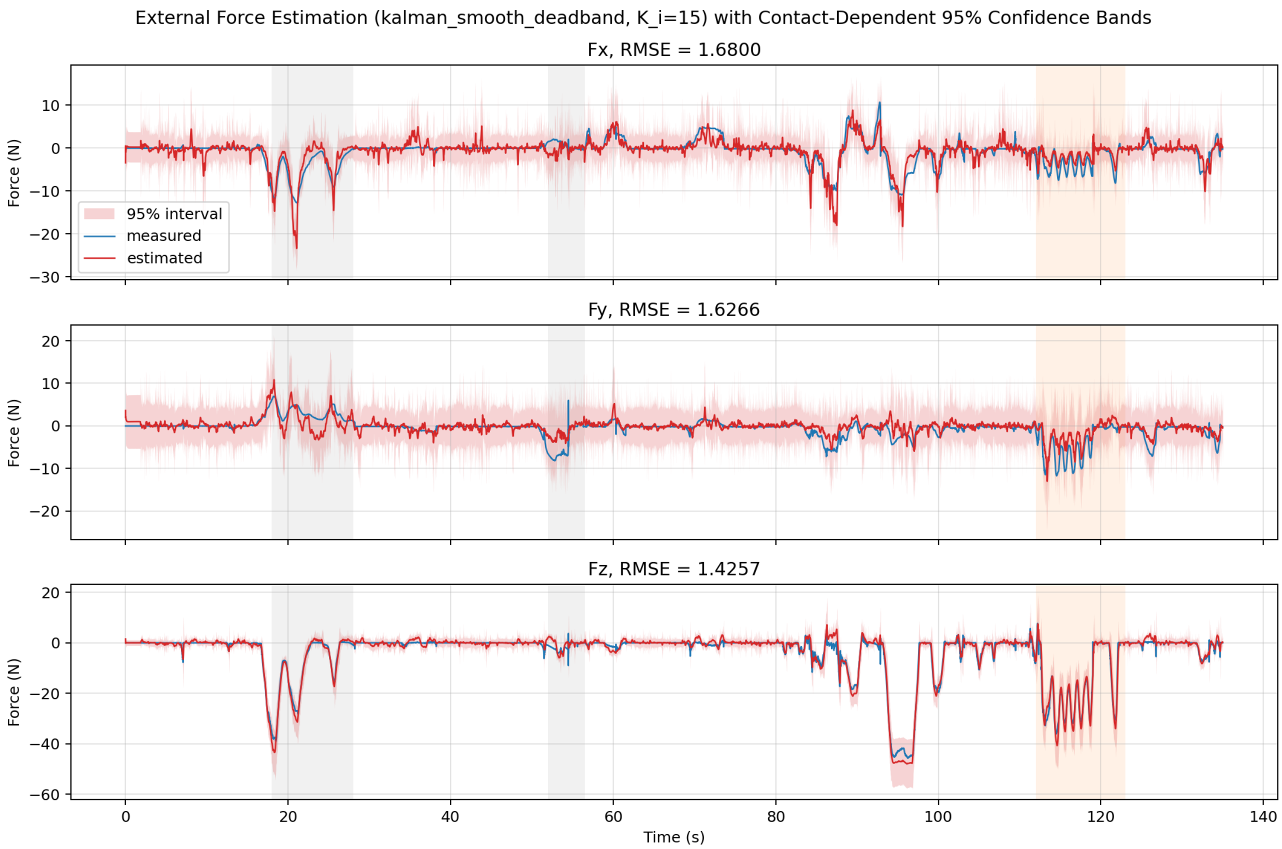

Figure 1: Estimated force, force-sensor validation, and leak-free 95% confidence bands for the active estimator. The force sensor validates performance but is not used to compute the band.

This plot answers whether the estimated force trajectory is plausible and whether the CI captures most validation force samples. Not every point must be covered by a 95% interval. If whole contact events are outside the band, the centerline estimator or uncertainty model needs improvement.

7.2 CI Width Versus Estimated Contact Force

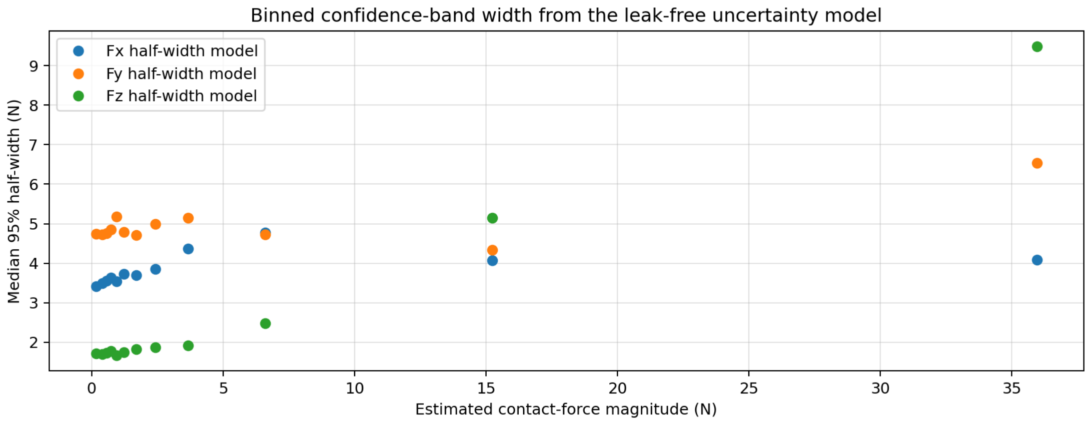

Figure 2: Binned median CI half-width versus estimated force magnitude. This is a diagnostic plot, not a forced monotonic law.

The original expectation was that uncertainty should shrink as force grows. The current model is more nuanced:

- Near zero force, uncertainty is high because contact/no-contact is ambiguous.
- At moderate force, uncertainty can narrow.
- At large force, uncertainty may widen because relative calibration/model error grows with force magnitude.

Therefore, a mild U-shape or increasing high-force tail is expected when the relative uncertainty term is active.

7.3 Velocity Context for Force Uncertainty

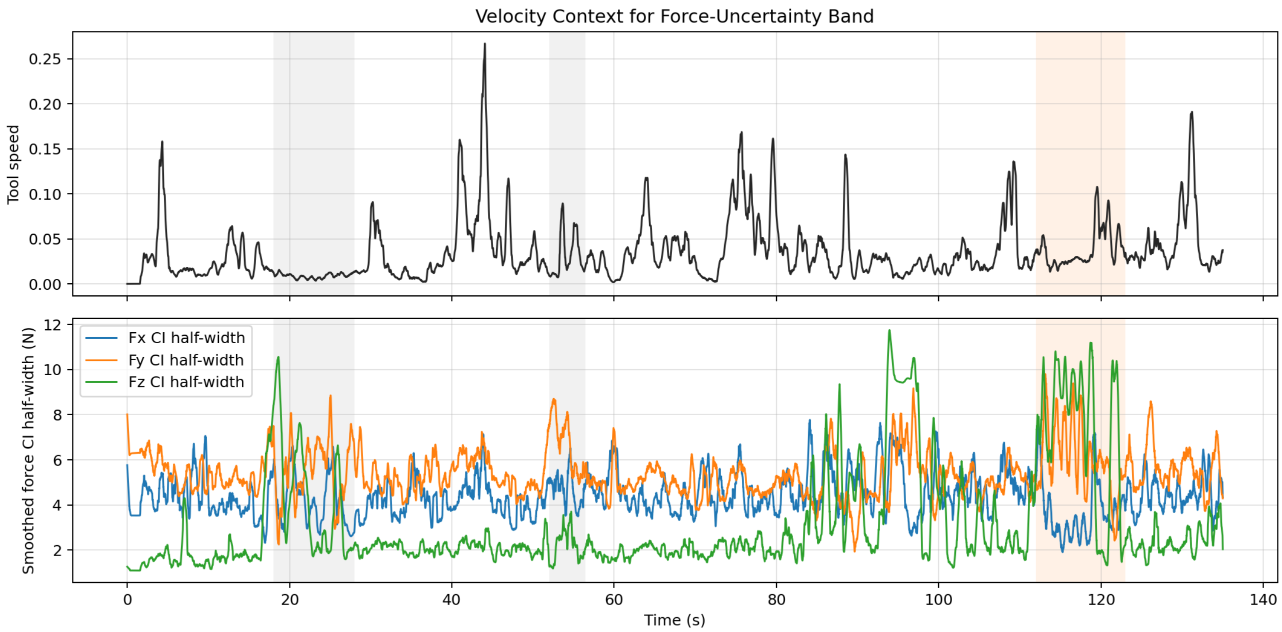

Figure 3: Tool speed from $J\dot{q}$ shown with smoothed force CI half-widths. This is a velocity-context plot, not a velocity confidence interval.

This plot helps address the advisor's concern: whether the force uncertainty is only a proxy for velocity. The conclusion from the current data is that velocity does affect some transient regions, especially fast teleoperation, but force CI width is not explained by velocity alone.

7.4 CI Width Versus Tool Speed

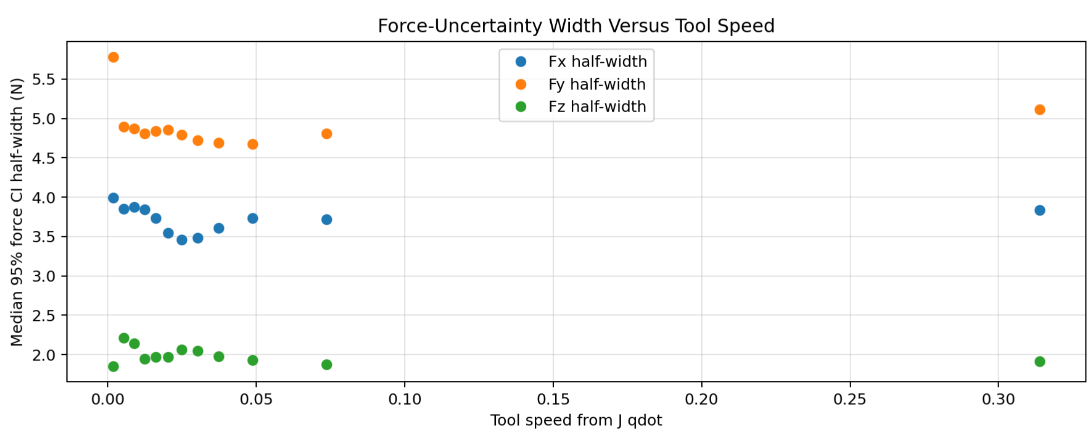

Figure 4: Binned median force CI half-width versus tool speed from $J\dot{q}$.

If force uncertainty were purely velocity-driven, this plot would show a strong monotonic increase. Instead, the relationship is weak and channel-dependent. This suggests that force uncertainty comes from several sources: torque residual ambiguity, filtering disagreement, Jacobian projection, and axis-dependent model/calibration error.

8 Main Conclusions

1. The force-estimation pipeline is realtime: torque residual filtering is causal, and force projection only requires the current Jacobian.
2. The Kalman filter is feasible and principled, but raw Kalman filtering is not enough because the residual contains false-positive/free-space effects.
3. The best Kalman-style method is smooth torque deadband followed by the Kalman filter.
4. The uncertainty band is leak-free because it never uses force-sensor values during CI computation.
5. The CI is not simply a velocity uncertainty band. Velocity provides useful context, but the force uncertainty is also driven by model/projection and filter-disagreement terms.
6. A true velocity confidence interval would require a separate velocity noise model or repeated velocity measurements. The current velocity plots are diagnostics to disentangle velocity from force-estimation uncertainty.